

CSCE 763: Digital Image Processing

Homework #2

Instructor: Spring 2026
University of South Carolina
Solutions by: JC Vaught

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Problem 1: Median Operator Nonlinearity (10 pts)

The median ζ , of a set of numbers is such that half the values in the set are below ζ , and the other half are above it. For example, the median of the set of values $\{12, 3, 8, 20, 21, 75, 31\}$ is 20. Show that an operator that computes the median of a subimage area, S , is nonlinear.

Hints: an operator H is linear if

$$H[a_1f_1(x, y) + a_2f_2(x, y)] = a_1H[f_1(x, y)] + a_2H[f_2(x, y)]$$

Step 1: State the Linearity Condition

An operator H is linear if and only if the **superposition property** holds for all images f_1, f_2 and all scalars a_1, a_2 :

$$H[a_1f_1 + a_2f_2] = a_1H[f_1] + a_2H[f_2]$$

To show H is nonlinear, it suffices to find a single counterexample where this equality fails. We set $a_1 = a_2 = 1$ and test the additivity property over a 1×3 subimage region S .

Step 2: Define the Counterexample

Choose two images defined by their pixel values over S :

$$f_1 = \{2, 3, 10\}, \quad f_2 = \{10, 1, 2\}$$

Step 3: Compute the Right-Hand Side

$$H[f_1] + H[f_2] = \text{median}\{2, 3, 10\} + \text{median}\{10, 1, 2\} = 3 + 2 = 5$$

Step 4: Compute the Left-Hand Side

$$f_1 + f_2 = \{2 + 10, 3 + 1, 10 + 2\} = \{12, 4, 12\}$$

$$H[f_1 + f_2] = \text{median}\{12, 4, 12\} = 12$$

Step 5: Conclude

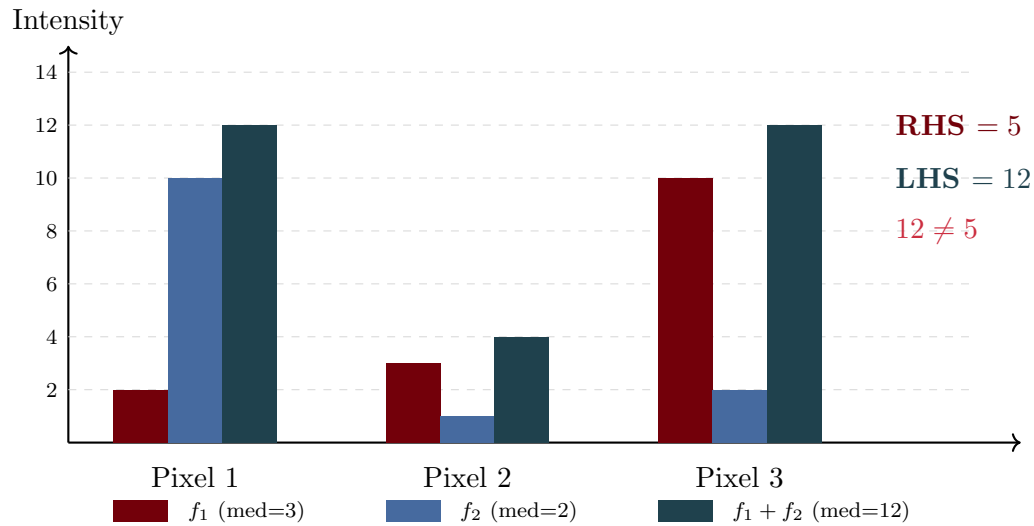
Since

$$H[f_1 + f_2] = 12 \neq 5 = H[f_1] + H[f_2]$$

the superposition property is violated. Therefore, the median operator is **nonlinear**.

■

Visualization: Counterexample Bar Chart



Results

The median operator is **nonlinear**. The counterexample $f_1 = \{2, 3, 10\}$, $f_2 = \{10, 1, 2\}$ yields:

$$H[f_1 + f_2] = 12 \neq 5 = H[f_1] + H[f_2] \quad \blacksquare$$

Problem 2: Image Subtraction for Defect Detection (20 pts)

Image subtraction is used often in industrial applications for detecting missing components in product assembly. The approach is to store a “golden” image that corresponds to a correct assembly; this image is then subtracted from incoming images of the same product. Ideally, the differences would be zero if the new products are assembled correctly. Difference images for products with missing components would be nonzero in the area where they differ from the golden image. What conditions do you think have to be met in practice for this method to work? You need to list at least two major factors that affect the detection performance.

Step 1: Consistent Illumination

The lighting conditions—direction, intensity, color temperature, and uniformity—must be **identical** when capturing the golden image and every subsequent test image. Any change in illumination causes intensity differences unrelated to actual defects, producing false positives. In practice, this requires fixed, controlled lighting enclosures.

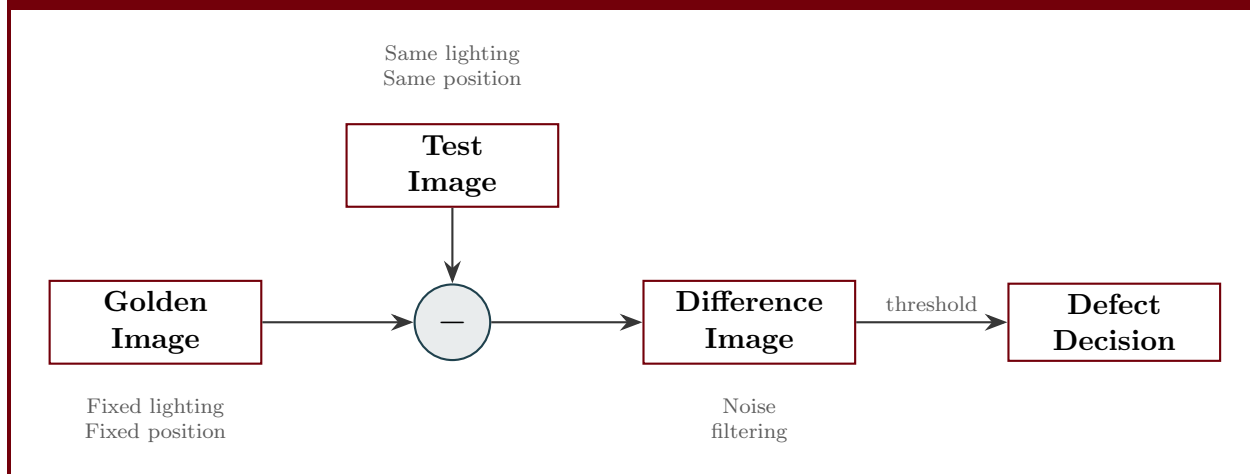
Step 2: Precise Spatial Registration

The camera position, orientation, and product placement must be **exactly the same** for every image. Even sub-pixel misalignment between the golden and test images produces nonzero differences at edges and high-contrast boundaries, masking true defects or creating false detections. Mechanical fixtures and calibrated camera mounts achieve this.

Step 3: Additional Factors

- **Sensor noise thresholding:** Random sensor noise produces small nonzero differences even with perfect alignment and lighting. A threshold on the difference image distinguishes genuine defects from noise artifacts.
- **Consistent camera settings:** Gain, exposure time, white balance, and focus must remain fixed between the golden capture and all subsequent captures.

Visualization: Image Subtraction Flowchart



Results

The two major factors are: (1) **consistent illumination** so that intensity differences reflect actual physical differences, not lighting variations; and (2) **precise spatial alignment/registration** so that geometric misalignment does not produce spurious nonzero differences.

Problem 3: Histogram Equalization Idempotency (20 pts)

Suppose that a digital image is subjected to histogram equalization. Show that a second pass of histogram equalization (on the histogram-equalized image) will produce exactly the same result as the first pass.

Step 1: Define the Equalization Transform

Let the original image have intensity levels $r \in [0, L - 1]$ with PDF $p_r(r)$ and CDF $P_r(r)$. The histogram equalization transformation is:

$$s = T(r) = (L - 1) P_r(r) = (L - 1) \int_0^r p_r(w) dw$$

Step 2: Show the Equalized Image Is Uniform

A fundamental result from probability theory: if $s = T(r) = (L - 1) P_r(r)$, then s is **uniformly distributed** on $[0, L - 1]$:

$$p_s(s) = \frac{1}{L - 1}, \quad 0 \leq s \leq L - 1$$

Step 3: Apply Equalization Again

Apply histogram equalization a second time to the equalized image:

$$v = T_2(s) = (L - 1) P_s(s) = (L - 1) \int_0^s p_s(w) dw$$

Since $p_s(s) = \frac{1}{L - 1}$, the CDF is:

$$P_s(s) = \int_0^s \frac{1}{L - 1} dw = \frac{s}{L - 1}$$

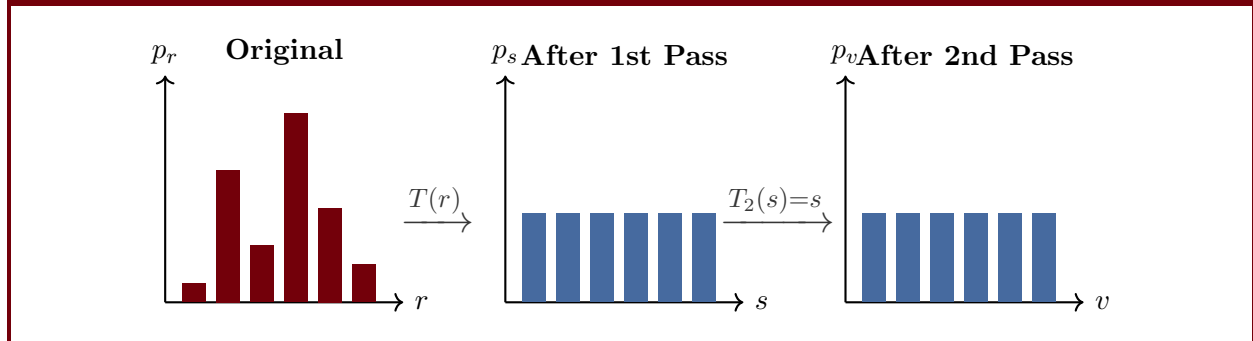
Step 4: Simplify to Identity

Substituting:

$$v = T_2(s) = (L - 1) \cdot \frac{s}{L - 1} = s$$

The second equalization is the **identity transformation**: $v = s$. The output is unchanged. ■

Visualization: Histogram Through Two Equalization Passes



Results

After the first histogram equalization, the output has a uniform distribution. Applying histogram equalization to a uniformly distributed image yields the identity mapping $v = s$. Therefore, the second pass produces exactly the same result as the first. ■

Problem 4: Histogram Matching (30 pts)

Suppose that a 4-bit image ($L = 16$) of size 64×64 has the intensity distribution as Table A. It is desired to transform the original histogram to a specified histogram as shown in Table B. What will be the actual histogram after the histogram matching operation?

r_k	n_k
0	16
1	0
2	512
3	32
4	64
5	128
6	64
7	1024
8	512
9	512
10	16
11	1024
12	0
13	128
14	0
15	64

Table A

z_q	n_q
0	64
1	128
2	512
3	512
4	1024
5	1024
6	512
7	128
8	128
9	64
10	0
11	0
12	0
13	0
14	0
15	0

Table B
Step 1: Setup

Total pixels: $N = 64 \times 64 = 4096$, intensity levels: $L = 16$, so $L - 1 = 15$.
The equalization transform is:

$$s_k = \text{round} \left(15 \cdot \sum_{j=0}^k \frac{n_j}{4096} \right)$$

Step 2: Cumulative Histogram and s_k Values from Table A

r_k	n_k	$\sum_{j=0}^k n_j$	$s_k = \text{round}\left(\frac{15 \cdot \sum_{j=0}^k n_j}{4096}\right)$
0	16	16	$\text{round}(0.059) = 0$
1	0	16	$\text{round}(0.059) = 0$
2	512	528	$\text{round}(1.934) = 2$
3	32	560	$\text{round}(2.051) = 2$
4	64	624	$\text{round}(2.285) = 2$
5	128	752	$\text{round}(2.754) = 3$
6	64	816	$\text{round}(2.988) = 3$
7	1024	1840	$\text{round}(6.738) = 7$
8	512	2352	$\text{round}(8.613) = 9$
9	512	2864	$\text{round}(10.488) = 10$
10	16	2880	$\text{round}(10.547) = 11$
11	1024	3904	$\text{round}(14.297) = 14$
12	0	3904	$\text{round}(14.297) = 14$
13	128	4032	$\text{round}(14.766) = 15$
14	0	4032	$\text{round}(14.766) = 15$
15	64	4096	$\text{round}(15.000) = 15$

Step 3: Cumulative Histogram and $G(z_q)$ Values from Table B

z_q	n_q	$\sum_{j=0}^q n_j$	$G(z_q) = \text{round}\left(\frac{15 \cdot \sum_{j=0}^q n_j}{4096}\right)$
0	64	64	$\text{round}(0.234) = 0$
1	128	192	$\text{round}(0.703) = 1$
2	512	704	$\text{round}(2.578) = 3$
3	512	1216	$\text{round}(4.453) = 4$
4	1024	2240	$\text{round}(8.203) = 8$
5	1024	3264	$\text{round}(11.953) = 12$
6	512	3776	$\text{round}(13.828) = 14$
7	128	3904	$\text{round}(14.297) = 14$
8	128	4032	$\text{round}(14.766) = 15$
9	64	4096	$\text{round}(15.000) = 15$
10	0	4096	15
11	0	4096	15
12	0	4096	15
13	0	4096	15
14	0	4096	15
15	0	4096	15

Step 4: Inverse Mapping $s_k \rightarrow z_q$

For each value of s_k , find the smallest z_q such that $G(z_q) \geq s_k$:

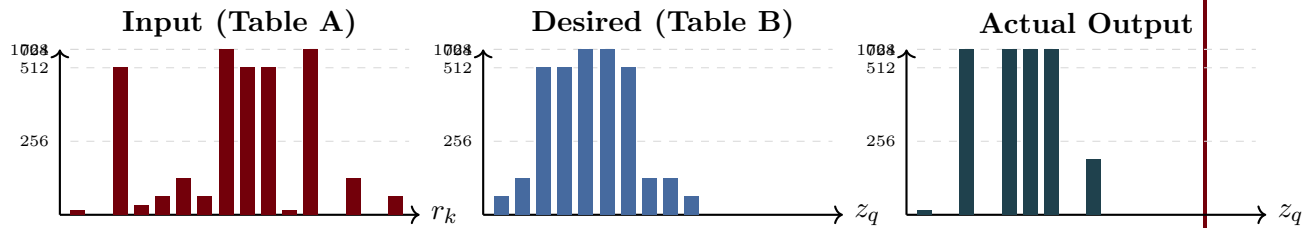
r_k	s_k	Smallest z_q with $G(z_q) \geq s_k$	z_q
0	0	$G(0) = 0 \geq 0$	0
1	0	$G(0) = 0 \geq 0$	0
2	2	$G(2) = 3 \geq 2$	2
3	2	$G(2) = 3 \geq 2$	2
4	2	$G(2) = 3 \geq 2$	2
5	3	$G(2) = 3 \geq 3$	2
6	3	$G(2) = 3 \geq 3$	2
7	7	$G(4) = 8 \geq 7$	4
8	9	$G(5) = 12 \geq 9$	5
9	10	$G(5) = 12 \geq 10$	5
10	11	$G(5) = 12 \geq 11$	5
11	14	$G(6) = 14 \geq 14$	6
12	14	$G(6) = 14 \geq 14$	6
13	15	$G(8) = 15 \geq 15$	8
14	15	$G(8) = 15 \geq 15$	8
15	15	$G(8) = 15 \geq 15$	8

Step 5: Build the Output Histogram

Using the mapping $r_k \rightarrow z_q$ and the pixel counts n_k from Table A:

Output z_q	Mapped from r_k	Pixel count
0	r_0, r_1	$16 + 0 = 16$
1	—	0
2	r_2, r_3, r_4, r_5, r_6	$512 + 32 + 64 + 128 + 64 = 800$
3	—	0
4	r_7	1024
5	r_8, r_9, r_{10}	$512 + 512 + 16 = 1040$
6	r_{11}, r_{12}	$1024 + 0 = 1024$
7	—	0
8	r_{13}, r_{14}, r_{15}	$128 + 0 + 64 = 192$
9–15	—	0 each
Total:		$16 + 800 + 1024 + 1040 + 1024 + 192 = 4096 \checkmark$

Visualization: Input



Results

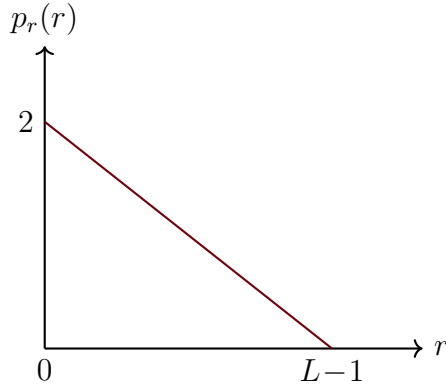
The actual output histogram after histogram matching is:

z_q	n_q^{out}
0	16
1	0
2	800
3	0
4	1024
5	1040
6	1024
7	0
8	192
9	0
10	0
11	0
12	0
13	0
14	0
15	0

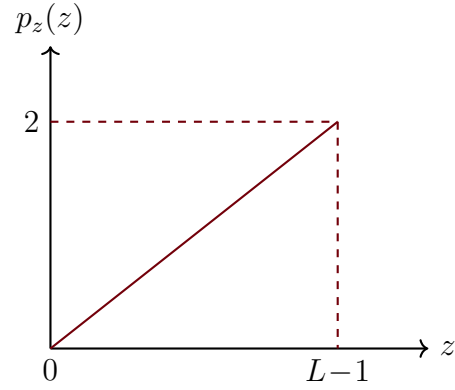
The output histogram approximates the desired distribution but is not identical to it, due to the discrete nature of the intensity levels and the many-to-one mapping inherent in histogram matching.

Problem 5: Continuous Intensity Transformation (20 pts)

An image with intensities in the range $[0, L - 1]$ has the pdf $p_r(r)$ shown in Figure (a). It is desired to transform the intensity levels of this image so that they will have the specified pdf $p_z(z)$ as shown in Figure (b). Assume continuous quantities and find the transformation $z = f(r)$ that will accomplish this.



(a)



(b)

Step 1: Write $p_r(r)$ with Normalization Verification

Let $W = L - 1$. From Figure (a), $p_r(r)$ is a line from $(0, \frac{2}{W})$ to $(W, 0)$:

$$p_r(r) = \frac{2}{W} \left(1 - \frac{r}{W} \right) = \frac{2(W - r)}{W^2}, \quad 0 \leq r \leq W$$

$$\text{Verification: } \int_0^W \frac{2(W - r)}{W^2} dr = \frac{2}{W^2} \left[Wr - \frac{r^2}{2} \right]_0^W = \frac{2}{W^2} \cdot \frac{W^2}{2} = 1. \checkmark$$

Step 2: Write $p_z(z)$ with Normalization Verification

From Figure (b), $p_z(z)$ is a line from $(0, 0)$ to $(W, \frac{2}{W})$:

$$p_z(z) = \frac{2z}{W^2}, \quad 0 \leq z \leq W$$

$$\text{Verification: } \int_0^W \frac{2z}{W^2} dz = \frac{2}{W^2} \cdot \frac{W^2}{2} = 1. \checkmark$$

Step 3: Compute CDF $T(r)$

$$T(r) = \int_0^r p_r(w) dw = \int_0^r \frac{2(W-w)}{W^2} dw = \frac{2}{W^2} \left[Wr - \frac{r^2}{2} \right] = \frac{2Wr - r^2}{W^2}$$

Step 4: Compute CDF $G(z)$

$$G(z) = \int_0^z p_z(w) dw = \int_0^z \frac{2w}{W^2} dw = \frac{z^2}{W^2}$$

Step 5: Set $G(z)$

$$\frac{z^2}{W^2} = \frac{2Wr - r^2}{W^2}$$

$$z^2 = 2Wr - r^2 = r(2W - r)$$

$$z = \sqrt{r(2(L-1) - r)}$$

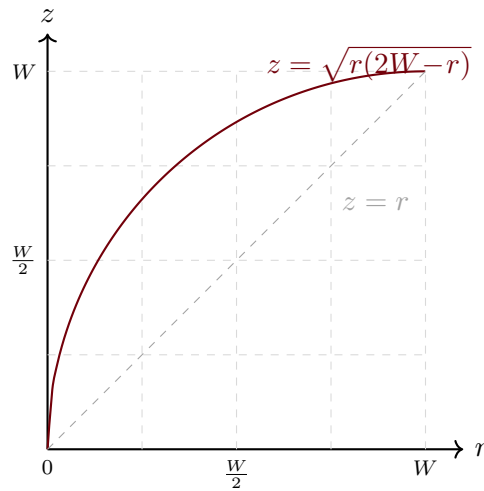
where we take the positive root since $0 \leq r \leq W$ ensures the argument is non-negative.

Step 6: Verify Boundary Conditions

- $f(0) = \sqrt{0 \cdot (2W - 0)} = 0 \checkmark$
- $f(W) = \sqrt{W(2W - W)} = \sqrt{W^2} = W = L - 1 \checkmark$

The transformation maps $[0, L - 1] \rightarrow [0, L - 1]$ as required.

Visualization: Transformation z



The curve is a quarter-circle arc from $(0,0)$ to (W,W) , lying above the identity line, indicating the transformation boosts lower intensities.

Results

The transformation that maps the input PDF $p_r(r)$ to the desired output PDF $p_z(z)$ is:

$$z = f(r) = \sqrt{r(2(L-1) - r)}, \quad 0 \leq r \leq L-1$$

This is a quarter-circle arc from $(0,0)$ to $(L-1, L-1)$. ■